ANDERSON JUNIOR COLLEGE

2017 JC 2 PRELIMINARY EXAMINATIONS

NAME:_________________________ **PDG:**____ /16

CHEMISTRY **9729/04**

Paper 4 Practical

23 August 2017

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.
Give details of the practical shift and laboratory where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Quantitative Analysis Notes are printed on pages 17 and 18.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **18** printed pages.

Answer **all** the questions in the spaces provided.

1 Determination of the concentration of thiosulfate ions

Potassium manganate(VII), $KMnO_4$, is an oxidising agent that can be used to determine the concentration of a solution of sodium thiosulfate.

To do this, aqueous potassium manganate(VII) is used to react with excess aqueous potassium iodide, KI to produce iodine solution. In this reaction, iodide ions are oxidised to iodine by manganate(VII) ions in acidic solution.

$$2MnO_4^-(aq) + 10I^-(aq) + 16H^+(aq) \longrightarrow 2Mn^{2+}(aq) + 5I_2(aq) + 8H_2O(l)$$

The iodine produced is then titrated with aqueous thiosulfate ions.

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

You are provided with the following.
FA 1 is 0.100 mol dm^{-3} of potassium manganate(VII), $KMnO_4$.
FA 2 is an aqueous solution of sodium thiosulfate, $Na_2S_2O_3$.
FA 3 is an aqueous solution of potassium iodide, KI.
FA 4 is dilute sulfuric acid, H_2SO_4.

In this question, you are to perform a dilution of the $KMnO_4$ provided before carrying out a titration. The data from this titration will be used to determine the concentration of $S_2O_3^{2-}$ in **FA 2**.

(a) (i) Dilution of FA 1

1. Fill a burette with **FA 1**.
2. Run between 40.00 cm^3 and 45.00 cm^3 of **FA 1** into 250 cm^3 volumetric flask.
3. Make up to the mark with deionised water.
4. Shake the volumetric flask to obtain a homogeneous solution and label this solution **FA 5**.
5. Record all burette readings and the volume of **FA 1** that you used for dilution in the space provided in Table 1.1 below.

Table 1.1

Final burette reading / cm^3	
Initial burette reading / cm^3	
Volume of **FA 1** used / cm^3	

[1]

(ii) Titration of iodine liberated by FA 5 with FA 2

1. Fill a second burette with **FA 2**.
2. Using a pipette, transfer 25.0 cm^3 of **FA 5** into the conical flask.
3. Using appropriate measuring cylinders, add about 10 cm^3 of **FA 3** and about 10 cm^3 of **FA 4** to the conical flask.
4. Run **FA 2** from the burette into this flask until the colour of the solution just turns colourless.
5. Record your titration results in the space provided below. Make certain that your recorded results show the precision of your working.
6. Repeat points **1** to **5** as necessary until consistent results are obtained.

Results

[6]

(iii) From your titrations, obtain a suitable volume of **FA 2** to be used in your calculations. Show clearly how you obtained this volume.

volume of **FA 2** = ………………………………………….. [1]

(b) **(i)** Calculate the amount of manganate(VII) ions, MnO_4^-, in 25.0 cm^3 of **FA 5**.

amount of MnO_4^- in 25.0 cm^3 of **FA 5** = ……………………………………….. [2]

(ii) Given that one mole of manganate(VII) ions liberates sufficient iodine from potassium iodide to react with five moles of thiosulfate ions, $S_2O_3^{2-}$, calculate the number of moles of thiosulfate ions present in the volume of **FA 5** that you found necessary in the titration.

amount of $S_2O_3^{2-}$ ions needed = ……………………………………….. [1]

(iii) Determine the concentration, in mol dm^{-3}, of $S_2O_3^{2-}$ in **FA 2**.

concentration of $S_2O_3^{2-}$ in **FA 2** = ……………………………………….. [1]

(c) A student suggests that the volumes of **FA 3** and **FA 4** should be measured using a burette instead of measuring cylinders, so as to improve the accuracy of the titration data.

Do you agree with the student? Explain your answer.

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(d) A teacher performs this same experiment, using the quantities described earlier, and obtains a mean titre volume of 25.65 cm^3.

The errors (uncertainties) associated with each reading using a volumetric flask, pipette and burette are ±0.15 cm^3, ±0.1 cm^3 and ±0.05 cm^3 respectively.

Calculate the maximum total percentage error (uncertainty) of this mean titre value.

[1]

(e) Planning

X is another oxidising agent that can react with potassium iodide, KI, to give iodine, I_2.

The rate equation for this reaction is rate = k [KI]^a [X]^b, where a and b are the orders of reaction with respect to KI and X respectively.

An investigation was carried out to determine the order of reaction with respect to KI. This is done by mixing different volumes of the X and KI, together with a fixed volume of sodium thiosulfate, $Na_2S_2O_3$. The I_2 produced from the reaction between X and KI will react with the $Na_2S_2O_3$ added and be reduced back to iodide, I^-. When all the $Na_2S_2O_3$ is used up, the colour of I_2 will appear.

The kinetics of this reaction can be determined by measuring the time taken for the colour of I_2 to first appear.

The following experiments were carried out to determine the order of reaction with respect to KI.

Experiment	volume of X / cm^3	volume of KI / cm^3	volume of H_2O / cm^3	volume of $Na_2S_2O_3$ / cm^3	time / s
1	50	40	0	10	40
2	50	20	20	10	82

(i) From the data given above, state the order of reaction with respect to KI.

…………………………………………………………………………………………………. [1]

(ii) Explain clearly how the above results support your answer in **(e)(i)**.

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(iii) Sketch the graph of concentration of KI against time based on the order you have determined in **(e)(i).**

[1]

(iv) You are to plan an investigation for **one further experiment**, which together with those above, will allow you to determine the order of reaction with respect to X.

In your plan, you should include details of
- the volumes of the solutions you would use
- the apparatus and their capacities that you would use
- the procedure that you would follow, including details on how you would obtain reliable results.

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[Total: 23]

2 **Determination of the enthalpy change of a reaction, ΔH_r**

You are required to obtain values of ΔH for two chemical reactions and use them to calculate ΔH for a third reaction.

FA 6 is 1 mol dm^{-3} hydrochloric acid, HCl.
FA 7 is solid sodium carbonate, Na_2CO_3.
FA 8 is solid sodium hydrogencarbonate, $NaHCO_3$.

Experiment 1

1. Weigh an empty weighing bottle.

2. Place between 4.50 g and 5.00 g of **FA 7** in the weighing bottle.

3. Place a dry Styrofoam cup inside a 250 cm^3 beaker.

4. Using a measuring cylinder, place 50 cm^3 of **FA 6** into the Styrofoam cup.

5. Place the lid onto the cup through the thermometer from the top. Stir the liquid in the cup with the thermometer and measure its temperature when it reaches a steady temperature. Read and record this temperature in your table.

6. Tip **cautiously** the contents of the weighing bottle into the acid in the Styrofoam cup, stir gently with the thermometer and record the **highest** temperature obtained in your table.

7. Calculate the change in temperature.

8. Reweigh the empty weighing bottle.

In an appropriate format in the space provided below, record all measurements of mass and temperature. Calculate the mass of **FA 7** used.

Results

(a) If 4.2 J are required to raise the temperature of 1 cm^3 of solution by 1 ^oC, calculate the amount of heat absorbed in **Experiment 1**.

[1]

(b) Calculate the number of moles of HC*l* added to the cup.

[1]

(c) Calculate the number of moles of Na_2CO_3 added to the cup.
[M_r (Na_2CO_3) = 106.0]

[1]

(d) Explain which reagent was added in excess.

………………………………………………………………………………………………………

……………………………………………………………………………………………………… [1]

(e) Calculate the enthalpy change ΔH_1 for the following reaction.

$$Na_2CO_3(s) + 2HCl(aq) \longrightarrow 2NaCl(aq) + H_2O(l) + CO_2(g) \qquad \text{---}\ \Delta H_1$$

[1]

Experiment 2

1. Weigh an empty weighing bottle.

2. Place between 3.50 g and 4.00 g of **FA 8** in the weighing bottle.

3. Place a dry Styrofoam cup inside a 250 cm^3 beaker.

4. Using a measuring cylinder, place 50 cm^3 of **FA 6** into the Styrofoam cup.

5. Place the lid onto the cup through the thermometer from the top. Stir the liquid in the cup with the thermometer and measure its temperature when it reaches a steady temperature. Read and record this temperature in your table.

6. Tip **cautiously** the contents of the weighing bottle into the acid in the Styrofoam cup, stir gently with the thermometer and record the **lowest** temperature obtained in your table.

7. Calculate the change in temperature.

8. Reweigh the empty weighing bottle.

In an appropriate format in the space provided below, record all measurements of mass and temperature. Calculate the mass of **FA 8** used.

Results

[5]

(f) If 4.2 J are required to raise the temperature of 1 cm^3 of solution by 1 ^oC, calculate the amount of heat absorbed in **Experiment 2**.

[1]

(g) Calculate the number of moles of $NaHCO_3$ added to the cup.
[M_r ($NaHCO_3$) = 84.0]

[1]

(h) Calculate the enthalpy change ΔH_2 for the following reaction.

$$NaHCO_3(s) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g) \quad \text{---} \ \Delta H_2$$

[1]

(i) Use your answers in **(e)** and **(h)** to calculate ΔH_3 for the following reaction.

$$2NaHCO_3(s) \longrightarrow Na_2CO_3(s) + H_2O(l) + CO_2(g) \quad \text{---} \ \Delta H_3$$

[3]

(j) A student decides to repeat the experiment with 2.0 mol dm^{-3} HC*l* in **Experiment 2** instead of the given concentration of 1.0 mol dm^{-3} as specified in the instructions.

This mistake may lead to differences in heat exchange and reaction rate.

Is there any other effect this mistake would have on the value of ΔH_3 that she has calculated?

effect on ΔH_3 ....

explanation ....

....

.... [1]

[Total: 17]

3 Inorganic Qualitative Analysis

(a) Carry out the following tests on **FA 9** and **FA 10** which contain cations and anions from the following list:
NH_4^+, Mg^{2+}, Al^{3+}, Ca^{2+}, Cr^{3+}, Mn^{2+}, Fe^{2+}, Fe^{3+}, Cu^{2+}, Zn^{2+}, CO_3^{2-}, NO_3^-, NO_2^-, SO_3^{2-}, SO_4^{2-}, Cl^-, Br^-, I^-.

Record your observations in the space provided in Table 3.1.

Your answers should include
- details of colour changes and precipitates formed.
- the names of gases evolved and details of the test used to identify each one.

Use a fresh 1 cm^3 sample of each solution for each test.

In all tests, the reagent should be added gradually until no further change is observed, with shaking after each addition.

Table 3.1

test	observations with **FA 9**	observations with **FA 10**
Add NaOH(aq), followed by a spatula load of aluminium powder.		
Add concentrated HC*l* dropwise. To the **FA 9** and concentrated HC*l* mixture, add deionised water.		

test	observations with **FA 9**	observations with **FA 10**
Add KI(aq). To the resulting solution, add $Na_2S_2O_3$(aq).		
Add 1 cm^3 dilute H_2SO_4 followed by 1 cm^3 of K_2CrO_4(aq). Divide into 2 portions. To one portion, add KI(aq) followed by 1 cm^3 of starch solution. To the other portion, add NaOH(aq).		

[4]

Summary

Use your observations to identify the following ions

FA 9 contains the cation …………………………… and anion ……………………………

FA 10 contains the cation ……………………………

State and explain all the evidence for your identification of the anion in **FA 9**.

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…………………………………………………………………………………………………… [2]

(b) State the type of reaction that occurs when

(i) concentrated HC*l* is added to **FA 9**.

……………………………………………………………………………………………… [1]

(ii) K_2CrO_4(aq) is added to **FA 10**.

……………………………………………………………………………………………… [1]

(c) Planning

FA 11 is a sample of white powder which contains **one** of the four oxides:

magnesium oxide (MgO), aluminium oxide (Al_2O_3), zinc oxide (ZnO), silicon dioxide (SiO_2)

(i) Fill in the spaces in Table 3.2 the expected solubility for each of the oxides, when mixed separately, in dilute HC*l* and aqueous NaOH.

Table 3.2

	dilute HC*l*	NaOH(aq)
MgO		
Al_2O_3		
SiO_2		
ZnO		

[2]

(ii) Using your expected observation in **(c)(i)**, suggest a plan using test tube reactions to identify the oxide present in **FA 11**.

In planning your tests you may use **only** the following reagents and no others.

Dilute hydrochloric acid, HCl
Aqueous sodium hydroxide, $NaOH(aq)$
Aqueous ammonia, $NH_3(aq)$

You may use a flow scheme or a table to chart the sequence of reactions to establish the identity of **FA 11**.

Your completed plan should include expected observations, deductions with explanation and/or formula of species present at **each** stage. Definite deductions may be made from tests where there appears to be no reaction.

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[Total: 15]

Qualitative Analysis Notes
[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	***reaction with***	
	NaOH(aq)	NH_3(aq)
aluminium, Al^{3+}(aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH_4^+(aq)	ammonia produced on heating	–
barium, Ba^{2+}(aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca^{2+}(aq)	white ppt. with high [Ca^{2+}(aq)]	no ppt.
chromium(III), Cr^{3+}(aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu^{2+}(aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe^{2+}(aq)	green ppt. turning brown on contact with air insoluble in excess	green ppt. turning brown on contact with air insoluble in excess
iron(III), Fe^{3+}(aq)	red-brown ppt. insoluble in excess	red–brown ppt. insoluble in excess
magnesium, Mg^{2+}(aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn^{2+}(aq)	off-white ppt. rapidly turning brown on contact with air insoluble in excess	off-white ppt. rapidly turning brown on contact with air insoluble in excess
zinc, Zn^{2+}(aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

ion	*reaction*
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, Cl^-(aq)	gives white ppt. with Ag^+(aq) (soluble in NH_3(aq))
bromide, Br^-(aq)	gives pale cream ppt. with Ag^+(aq) (partially soluble in NH_3(aq))
iodide, I^-(aq)	gives yellow ppt. with Ag^+(aq) (insoluble in NH_3(aq))
nitrate, NO_3^-(aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil
nitrite, NO_2^-(aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil NO liberated by dilute acids (colourless NO $\rightarrow$ (pale) brown NO_2 in air)
sulfate, SO_4^{2-}(aq)	gives white ppt. with Ba^{2+}(aq) (insoluble in excess dilute strong acids)
sulfite, SO_3^{2-}(aq)	SO_2 liberated on warming with dilute acids; gives white ppt. with Ba^{2+}(aq) (soluble in dilute strong acids)

(c) Tests for gases

gas	*test and test result*
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	“pops” with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

halogen	*colour of element*	*colour in aqueous solution*	*colour in hexane*
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple